

Test yourself

Q1: Hoy et al, p56, Q6

Simplify the following expressions

- a) $\frac{\sqrt{a^5}}{a^3}$
- b) a^2b^3/a^2b
- c) $bx_1^{b-1}x_2^c/cx_1^bx_2^{c-1}$
- d) $(x_1^{1/b}x_2^c)^{b/c}$

Q2: Hoy et al, p56, Q1b

Give equations and sketch graphs of the lines passing through (-2,2) and parallel to $y = 2-5x$.

Q3: Hoy et al, p192, Q3

Find the slope of each of the following production functions, $y = f(L)$. Graph the functions and their derivative functions. Give the economics significance of the functions. In each case $L > 0$.

- a) $y = 64 \cdot L^{1/4}$
- b) $y = 10L + 2L^{1/2}$
- c) $y = 5L^3$

Q4: Hoy et al, p56, Q2

In a class of 120 students, everyone would take two hamburgers if the price were zero and no one would buy hamburgers if the price were \$4 or more. Assume that the class demand curve is linear and give the equation. Explain what this implies about the demand for hamburgers where the price is \$3.99?

Q5: Hoy et al, p56, Q4

Total revenue is price x quantity sold. Show that the total revenue curve corresponding to the demand curve found in the previous question is a quadratic. Is it convex or concave? At what value does its maximum or minimum occur?

Q6: Hoy et al, p230, Q5

A profit maximizing firm has a total cost function

$$C = 0.5x^3 - 3x^2 + 6x + 50$$

And sells into a competitive market on which the price is \$1. What output should it produce? What would your answer be if the price were \$2?

Q7: Hoy et al, p464, Q4

Show that the production function $y = x_1^{1/2} \cdot x_2^{2/3}$ is homogenous to degree $7/6$.

Q8: Hoy et al, p464, Q5

For the Cobb Douglas production function $y = A \cdot x_1^a \cdot x_2^b$, $A > 0$ and $a, b > 0$ defined on $\mathbb{R}_{++}^2$, show that the Euler theorem applies so that:

$$f_1 x_1 + f_2 x_2 = k \cdot f(x_1, x_2)$$

Where k is the degree of homogeneity.

Q9: Hoy et al, p515, Q1

Solve the following constrained optimization and minimization problems, illustrating your answers with diagrams in each case. Confirm that each is an optimum:

- a) Max $y = 2x_1 + 3x_2$ subject to $2x_1^2 + 5x_2^2 = 10$
- b) Max $y = x_1^{0.25} \cdot x_2^{0.75}$ subject to $2x_1^2 + 5x_2^2 = 10$
- c) Min $y = 2x_1 + 4x_2$ subject to $x_1^{0.25} \cdot x_2^{0.75} = 10$

Q10

Show that the price elasticity of demand is equal to 1 for a Cobb Douglas demand function.

Solutions

Q1: Hoy et al, p56, Q6

Simplify the following expressions

e) $\frac{\sqrt{a^5}}{a^3}$

f) a^2b^3/a^2b

g) $bx_1^{b-1}x_2^c/cx_1^bx_2^{c-1}$

h) $(x_1^{1/b}x_2^c)^{b/c}$

$$\frac{\sqrt{a^5}}{a^3} = a^{\frac{5}{2}-3} = a^{-\frac{1}{2}} = \frac{1}{\sqrt{a}}$$

$$\frac{a^2b^3}{a^2b} = b^2$$

$$\frac{bx_1^{b-1}x_2^c}{cx_1^bx_2^{c-1}} = \frac{b \cdot x_2}{c \cdot x_1}$$

$$(x_1^{1/b}x_2^c)^{b/c} = x_1^{1/c}x_2^b$$

Q2: Hoy et al, p56, Q1b

Give equations and sketch graphs of the lines passing through (-2,2) and parallel to $y = 2-5x$.

$$Y = -8 - 5x$$

Q3: Hoy et al, p192, Q3

Find the slope of each of the following production functions, $y = f(L)$. Graph the functions and their derivative functions. Give the economics significance of the functions. In each case $L > 0$.

d) $y = 64 \cdot L^{1/4}$

e) $y = 10L + 2L^{1/2}$

f) $y = 5L^3$

$$\begin{aligned}
3. \quad (a) \quad & f(L) = 64L^{1/4} \Rightarrow f'(L) = 16L^{-3/4} \Rightarrow f''(L) = -12L^{-7/4} < 0 \\
(b) \quad & f(L) = 10L + 2L^{1/2} \Rightarrow f'(L) = 10 + L^{-1/2} \Rightarrow f''(L) = -(1/2)L^{-3/2} < 0 \\
(c) \quad & f(L) = 5L^3 \Rightarrow f'(L) = 15L^2 \Rightarrow f''(L) = 30L > 0
\end{aligned}$$

Q4: Hoy et al, p56, Q2

In a class of 120 students, everyone would take two hamburgers if the price were zero and no one would buy hamburgers if the price were \$4 or more. Assume that the class demand curve is linear and give the equation. Explain what this implies about the demand for hamburgers where the price is \$3.99?

$$Q = 240 - 60.P$$

$$Q = 240 - 60 \times 3.99 = 0.6$$

Q5: Hoy et al, p56, Q4

Total revenue is price x quantity sold. Show that the total revenue curve corresponding to the demand curve found in the previous question is a quadratic. Is it convex or concave? At what value does its maximum of minimum occur?

$$TR = P.Q$$

$$TR = 240.P - 60.P^2$$

$$\text{Max/min } dTR/dP = 0$$

$$\frac{dTR}{dP} = 240 - 120.P = 0$$

$$P = 2$$

Max/min? second derivative

$$\frac{dTR'}{dP} = -120$$

It's a maximum, second derivative is a negative, so concave

Q6: Hoy et al, p230, Q5

A profit maximizing firm has a total cost function

$$C = 0.5x^3 - 3x^2 + 6x + 50$$

And sells into a competitive market on which the price is \$1. What output should it produce? What would your answer be if the price were \$2?

For $p=\$1$ and $p=\$2$, the firm should produce zero output.

Q7: Hoy et al, p464, Q4

Show that the production function $y = x_1^{1/2} \cdot x_2^{2/3}$ is homogenous to degree 7/6.

Q8: Hoy et al, p464, Q5

For the Cobb Douglas production function $y = A \cdot x_1^a \cdot x_2^b$, $A>0$ and $a,b>0$ defined on $\mathbb{R}_{++}^2$, show that the euler theorem applies so that:

$$f_1 x_1 + f_2 x_2 = k \cdot f(x_1, x_2)$$

Where k is the degree of homogeneity.

5. Computing the partial derivatives f_1 and f_2 gives us

$$f_1 = \alpha A x_1^{\alpha-1} x_2^\beta \quad \text{and} \quad f_2 = \beta A x_1^\alpha x_2^{\beta-1}$$

and so

$$\begin{aligned} f_1 x_1 + f_2 x_2 &= \left(\alpha A x_1^{\alpha-1} x_2^\beta \right) x_1 + \left(\beta A x_1^\alpha x_2^{\beta-1} \right) x_2 \\ &= \alpha A x_1^\alpha x_2^\beta + \beta A x_1^\alpha x_2^\beta \\ &= (\alpha + \beta) A x_1^\alpha x_2^\beta \end{aligned}$$

Thus $f_1 x_1 + f_2 x_2 = k f(x_1, x_2)$, where $k = \alpha + \beta$, which is Euler's theorem, and if $\alpha + \beta = 1$, then $f_1 x_1 + f_2 x_2 = f(x_1, x_2)$.

Q9: Hoy et al, p515, Q1

Solve the following constrained optimization and minimization problems, illustrating your answers with diagrams in each case. Confirm that each is an optimum:

- d) Max $y = 2x_1 + 3x_2$ subject to $2x_1^2 + 5x_2^2 = 10$
- e) Max $y = x_1^{0.25} \cdot x_2^{0.75}$ subject to $2x_1^2 + 5x_2^2 = 10$
- f) Min $y = 2x_1 + 4x_2$ subject to $x_1^{0.25} \cdot x_2^{0.75} = 10$

$$\begin{aligned}
 1. \quad (a) \quad & \left(\frac{5}{3} \sqrt{\frac{18}{19}}, \sqrt{\frac{18}{19}} \right) \\
 (b) \quad & (7.38, 11.07) \\
 (c) \quad & (3, 4)
 \end{aligned}$$

Q10

Show that the price elasticity of demand is equal to 1 for a Cobb Douglas demand function.

Cobb Douglas demand function: $X = \frac{Y}{P} \cdot \left(\frac{a}{a+b} \right)$

$$\text{Elasticity} = \frac{\partial X}{\partial P} \cdot \frac{P}{X} = -Y \cdot \left(\frac{a}{a+b} \right) \cdot P^{-2} \cdot \frac{P}{\frac{Y}{P} \cdot \left(\frac{a}{a+b} \right)} = \frac{P^{-2}}{P^2} = 1$$